

1)

Given:

$$0 \text{ dBm} = 1 \text{ mW} = 10^{-3} \text{ W}$$

We know,

$$P(\text{dBm}) = 10 \log_{10} \left(\frac{P(\text{W})}{1 \text{ mW}} \right)$$

$$\text{or, } P(\text{dBm}) = 10 \log_{10} \left(\frac{10}{0.001} \right)$$

$$\text{or, } P(\text{dBm}) = 10 \log_{10} (10000)$$

$$\therefore P(\text{dBm}) = 10 \times 4 = 40 \text{ dBm.}$$

2)

Given,

Total bandwidth = 24 MHz

One simplex channel = 30 KHz

Two simplex channel = One duplex channel

Reuse factor (N) = 4

Traffic/user = 0.1 Erlang

(a)

Total no. of duplex channels = $\frac{24,000 \text{ KHz}}{2 \times 30 \text{ KHz}}$

$$= \frac{24,000}{60} \text{ KHz}$$

$$= 400 \text{ KHz}$$

$$= 400$$

$\therefore$ Total no. of channels is 400.

For a four-cell reuse pattern,

$$\text{channel/cell} = \frac{400}{4}$$

$$= 100$$

(b)
Assume, 90% of the available 100 channels

are used i.e. $100 \times 0.90 = 90$ Erlangs

$\therefore$ Offered traffic = approx. 90 Erlangs,

$\therefore$ Each user generates 0.1 Erlangs,

$$\text{Users} = \frac{90}{0.1}$$

$$= 900 \text{ users/cell}$$

(c)

For 100 channels and approx. 90 erlangs offered traffic, Erlang B gives,

$$P_b = \frac{A^N / N!}{\sum_{k=0}^N A^k / k!}$$

For, $N = 100$, $A = 90$

The blocking probability is approximately

$$P_b \approx 2.70\%$$

$\therefore$ Approximately 2.7% of calls are blocked at this traffic level.

(d)

With 120° sectoring, each cell is divided into three sectors with total channel per cell 100 and approximately

$$\frac{100}{3} \approx 33 \text{ channels per sector}$$

Using the same blocking probability of approximately 2.70%, Erlang B gives approximately 25.5 Erlangs per 33-channel sector. For three sectors,

(b)

$$A_{\text{total}} = 25.5 \times 3$$

$$A_{\text{total}} \approx 76.5 \text{ Erlangs}$$

$$\text{Users} = \frac{76.5}{0.1} = 765 \text{ users/cell}$$

(e)

$$\text{Area} = (50 \times 50) \text{ km}^2 = 25,000 \text{ km}^2$$

$$\text{Each cell covers: } 5 \text{ km}^2$$

$$\text{No. of cells} = \frac{25,000}{5} = 500$$

$$\text{For omnidirectional antennas} = 500 \times 900$$

$$= 450,000 \text{ subscribers}$$

(f)

Using approximately 765 users/cell:

$$500 \times 765 = 382,500 \text{ subscribers}$$

3)

$$\text{Total spectrum} = 20 \text{ MHz}$$

$$\text{Simplex channel} = 25 \text{ kHz}$$

$$\text{Reuse factor} = 4$$

(a)

A duplex channel requires two simplex channels:

$$2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

$$\therefore \frac{20 \text{ MHz}}{50 \text{ kHz}} = \frac{20,000}{50} = 400$$

$$= 400 \text{ duplex channel}$$

(b)

Channels per cell :

$$\frac{400}{4} = 100 \text{ channels per cell.}$$

4)

Given,

Number of cells = 84

Total channels = 300

Traffic / user = 0.04 Erlang

Blocking Probability = 1%.

$N = 4, 7, 12$

(a)

For $N = 4$,

$$C = \frac{300}{4} = 75 \text{ channels}$$

For 75 channels at 1% blocking:

$$A \approx 60.73 \text{ Erlangs}$$

For $N = 7$,

$$C = \frac{300}{7} = 42 \text{ channels}$$

$$A \approx 30.77 \text{ Erlangs}$$

For $N = 12$

$$C = \frac{300}{12} = 25 \text{ channels}$$

$$A \approx 16.2 \text{ Erlangs}$$

(b)

$$U = \frac{A}{0.04}$$

For $N=4$,

$$U = \frac{60.73}{0.04} \approx 1518$$

For 84 cells,

$$1518 \times 84 \approx 127,512 \text{ users}$$

For $N=7$

$$U = \frac{30.77}{0.04} \approx 769$$

$$769 \times 84 \approx 64,620 \text{ users}$$

For $N=12$

$$U = \frac{16.12}{0.04} \approx 403$$

$$403 \times 84 \approx 33,852 \text{ users}$$

5)

(a)

$$R = \frac{270.833}{8}$$

$$\therefore R = 33.85 \text{ kbps/user}$$

(b)

$$\text{Useful rate} = 33.854 - 10.1 = 23.754 \text{ kbps}$$

Efficiency:

$$\eta = \frac{23.754}{33.854} \times 100$$

$$\therefore \eta = 70.16\%$$

6)

(a)

Since, channels are numbered 1, 2, 3, ..., 20
∴ There are 20 channels.

(b)

channels 1 to 12 ~~and~~ are assigned to cell 1 to 12 respectively using FCA strategy. channels 13-20 can be kept as additional/overflow channel and assigned according to traffic requirements.

7)

Features of DECT are

- Digital communication - provides good voice quality.
- Cordless Operation - allows mobility within the coverage area.
- Low-power transmission - suitable for indoor applications.
- Multiple users - TDMA/TDD technology allows multiple users to share radio resources.
- Security - supports authentication and encryption mechanisms.
- Roaming - supports mobility between compatible base station.
- Good voice quality - data transmission reduces interference and noise.

DECT commonly operates in the 1880-1900 MHz frequency band in Europe.

8)

Given,

No. of cells = 32

Radius = 1.6 km

Total traffic channels = 336

Reuse factor = 7

(a)

We know,

$$A = \frac{3\sqrt{3}}{2} R^2 \quad (\text{For hexagon})$$

$$= \frac{3\sqrt{3}}{2} (1.6)^2$$

$$A \approx 6.651 \text{ km}^2$$

For 32 cells,

$$A_{\text{total}} = 6.651 \times 32$$

$$A_{\text{total}} \approx 212.84 \text{ km}^2$$

(b)

No. of concurrent calls =

$$\text{channels rate} = \frac{336}{7}$$

$$= 48 \text{ channels/cell}$$

(c)

Each channel can support one concurrent call:

$$48 \times 32 = 1536 \text{ concurrent calls}$$

9)

ISDN stands for Integrated Services Digital Network. It is a digital telecommunications technology used to carry voice, video and data over telephone networks.

There are two major types:

① BRI - Basic Rate Interface

It consists of two B channels and one D channel i.e. $2B+D$ where each B channel has 64 kbps and D channel has 16 kbps data rate. with the total of 144 kbps.

With framing/overhead, the line rate is commonly 192 kbps.

Uses:

- Small offices → Home users
- Digital telephone service
- Internet / data connections

② PRI - Primary Rate Interface

It provides many B channels plus one D channel at approximately 1.544 Mbps and 2.048 Mbps respectively.

Uses:

- Large business → PBX systems
- Voice communications → Voice conferencing
- High-capacity digital connections

10)

Given,

forward-link bandwidth = 20 MHz

Radio channels = 20 KHz

Speech channel per radio channels = 6

No. of radio channels = $\frac{20 \text{ MHz}}{20 \text{ KHz}}$

$$= \frac{20,000}{20} = 1000 \text{ radio channel}$$

Each radio supports 6 speech channels :

$$1000 \times 6 = 6000 \text{ simultaneous users}$$

11)

Ans: SS7 (Signaling System No. 7) is a telecommunications signaling system used to exchange control information between network elements.

• Features :

- Out-of-band signaling - signaling information is carried separately from the voice channel.
- Fast call setup - call-control information can be exchanged rapidly.
- Call routing - provides information needed to route calls.
- Call termination - helps release network resources after a call ends.
- Network management - supports monitoring and control of telecommunications networks.
- Error detection and recovery - provides mechanisms for reliable signaling.

• Call setup

When a caller initiates a call:

- 1) The originating switch receives the call request.
- 2) SS7 signaling messages are generated.
- 3) The destination information is transmitted through the signaling network.
- 4) Network switches determine the appropriate route.
- 5) The destination switch alerts the receiving subscriber.
- 6) A voice path is established.

• Routing

SS7 exchanges signaling information between switches and network databases. This allows the network to determine where a call should be sent.

• Call termination

When a user ends a call:

- 1) A release message is generated.
- 2) SS7 communicates the release information.
- 3) Switches disconnect the call.
- 4) Network resources become available for other call.

• Advantages

→ It uses a separate signaling network.

- It has faster call setup.
- It is more efficient.
- It has better security and control.
- It supports advanced services.
- Voice channel remains for voice in SS7.

12)

Given,

New calls per hour = 1800

Average holding time = 3 minutes

Traffic density:

$$A = \lambda H$$

converting holding time into hours,

$$H = \frac{3}{60} = 0.05 \text{ hour}$$

$$\therefore A = 1800 \times 0.05 \\ = 90 \text{ Erlangs}$$

13)

Given,

Total bandwidth = 33 MHz

Simplex channel = 25 kHz

$N = 4, 7, 12$

One duplex channel requires:

$$2 \times 25 = 50 \text{ kHz}$$

$$\therefore \text{Total duplex channel} = \frac{33 \text{ MHz}}{50 \text{ kHz}} = \frac{33000}{50} \\ = 660 \text{ duplex channel}$$

For $N = 4$,

$$\frac{660}{4} = 165 \text{ channels / cell}$$

For $N=7$,

$$\frac{660}{7} = 94.2857 \text{ channels / cell}$$

For $N=12$,

$$\frac{660}{12} = 55 \text{ channels / cell}$$

14)

Given $R = 48.6 \text{ kbps}$

No. of users = 3

Raw data per user :

$$R_u = \frac{48.6}{3}$$

$$= 16.2 \text{ kbps / user}$$